

KPP Validation Report

MITgcm vs Python Port Comparison

Generated: 2026-09-27 10:05:47

MITgcm Reference:

File: mitgcm_kpp_outputs_11k_1D.nc

UUID: 9ce2edbc-2bbc-4d49-b1b3-63d37cb07036

Python Port Output:

File: python_kpp_outputs_11k_1D.nc

UUID: 9ce2edbc-2bbc-4d49-b1b3-63d37cb07036

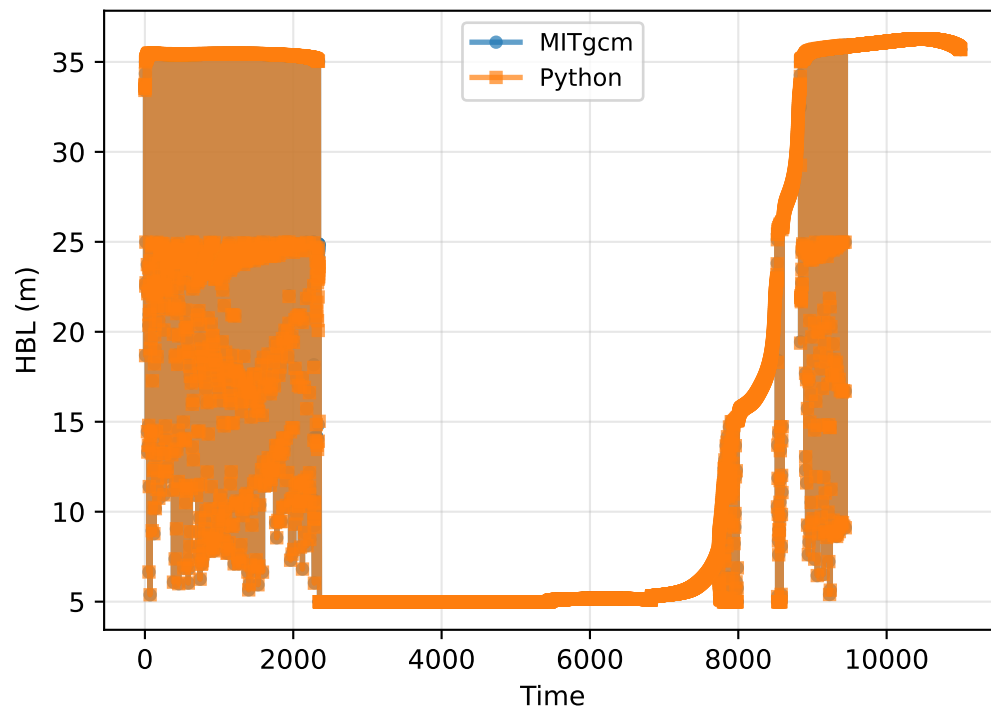
Provenance: ☐ UUIDs match

Timesteps: 11000

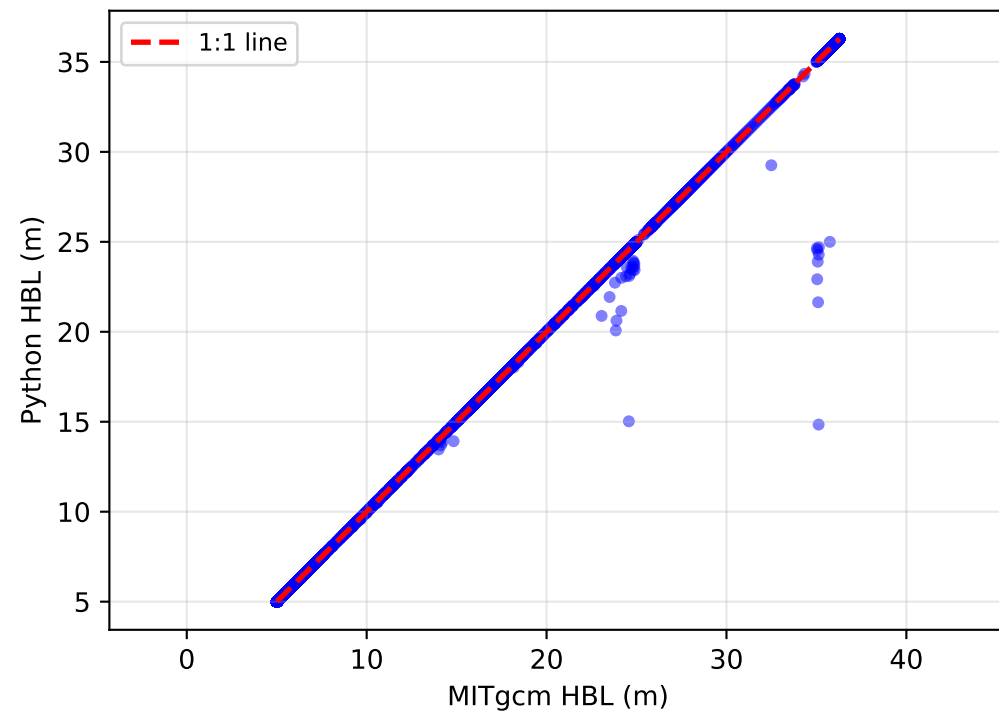
Grid: 1 × 1 × 23

Boundary Layer Depth (HBL) Comparison

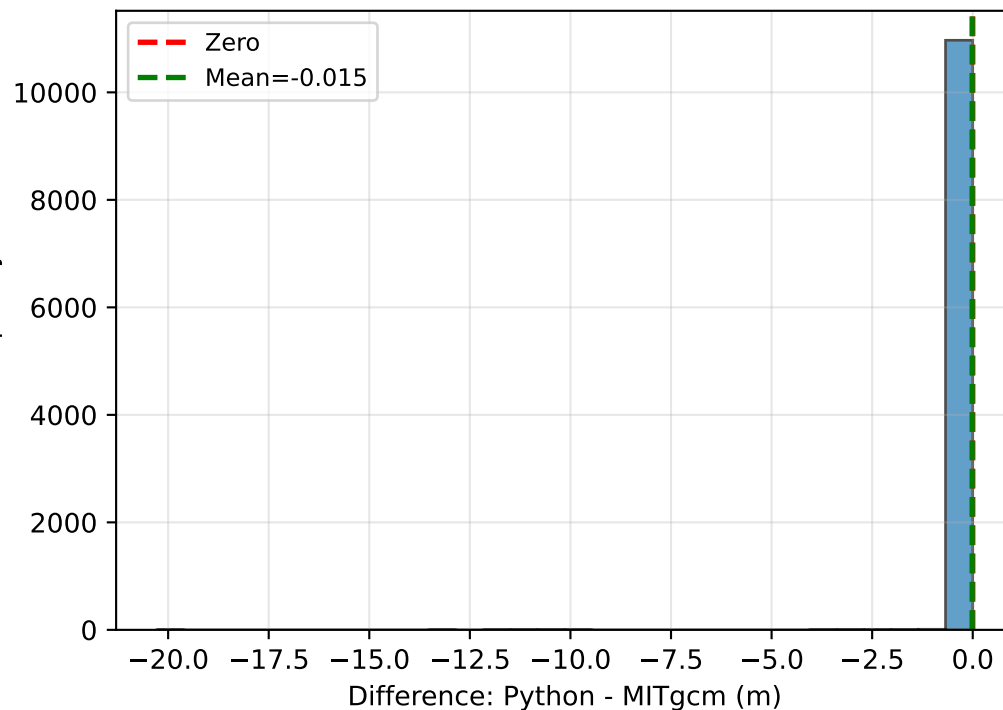
HBL Time Series



HBL Scatter Plot



HBL Difference Distribution



HBL Statistics:

Mean |diff|: 0.015214 m
Max |diff|: 20.279218 m
RMS diff: 0.380114 m

Mean rel error: 0.0604%
Median rel error: 0.0001%

Points compared: 11000

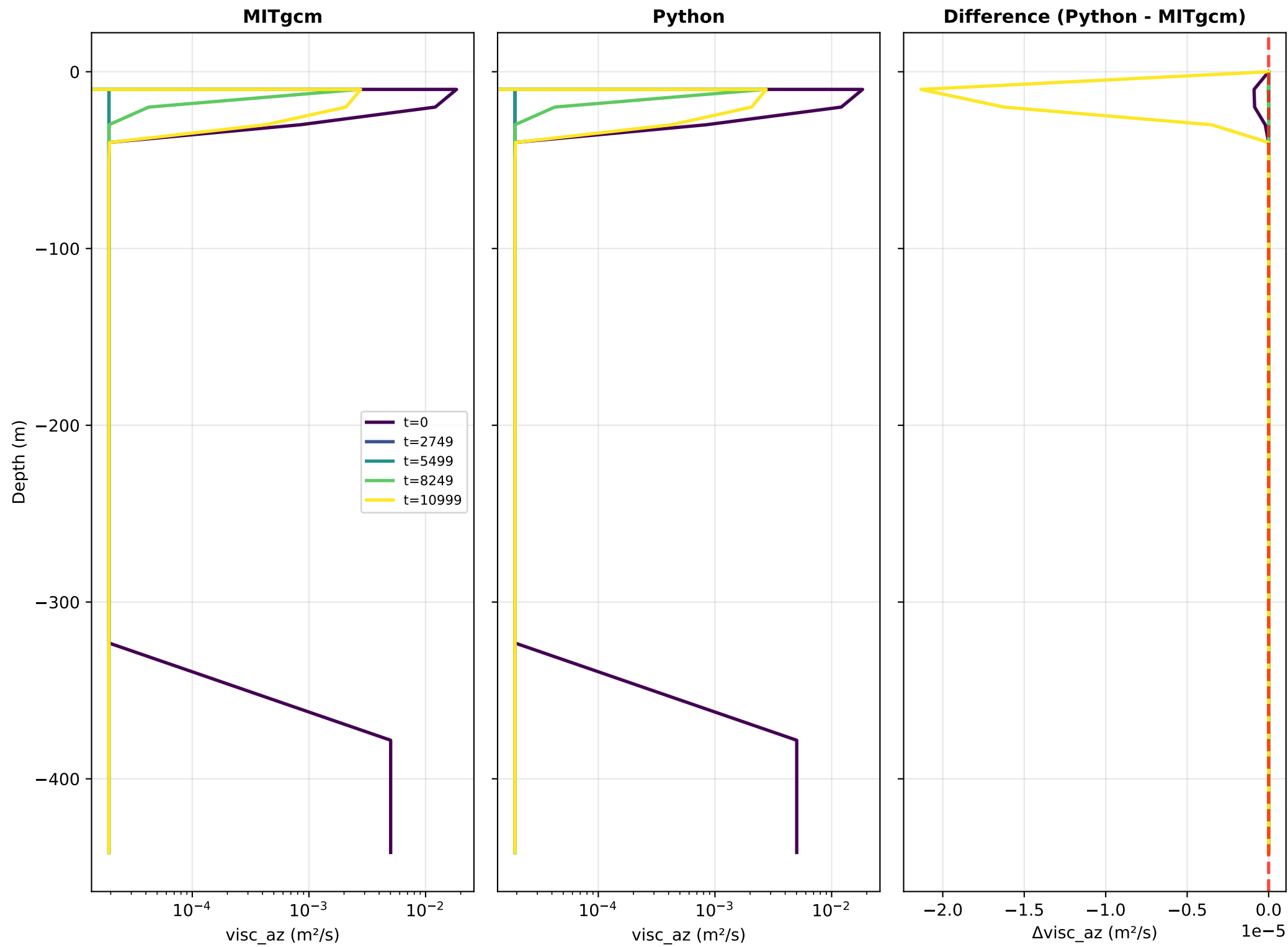
Range:

MITgcm: 5.0 – 36.3 m
Python: 5.0 – 36.3 m

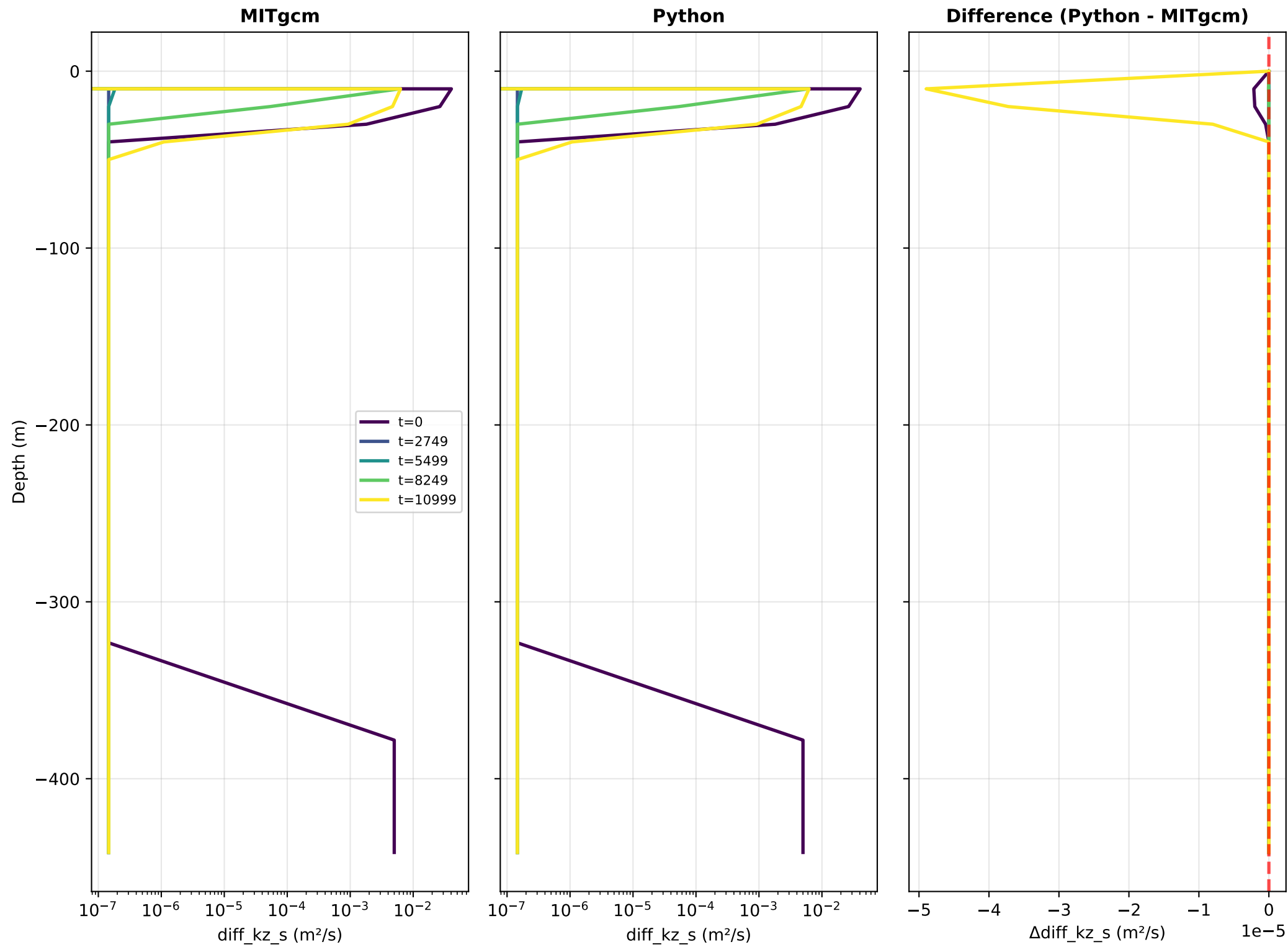
Assessment:

△ □ CHECK

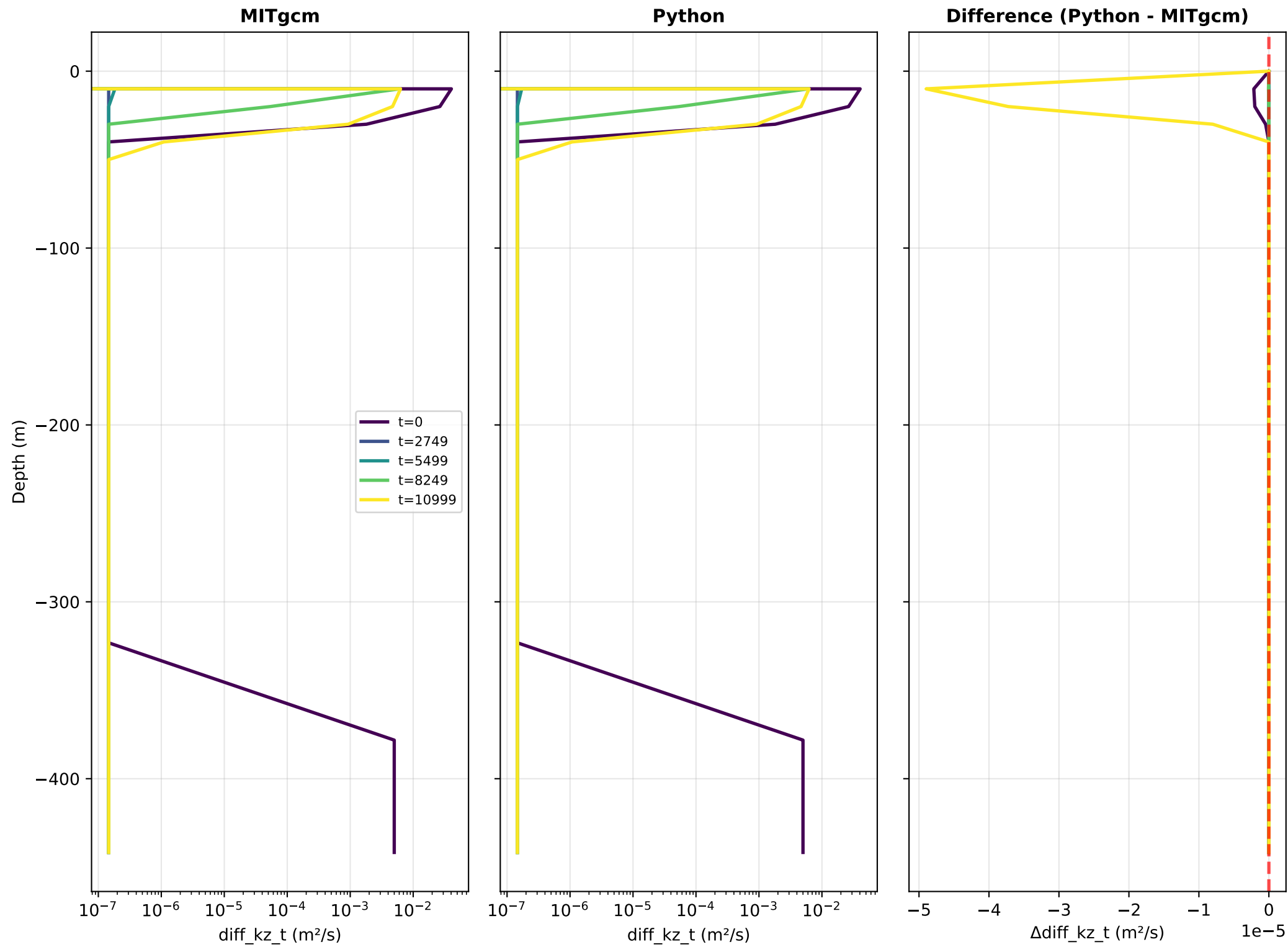
VISC_AZ Vertical Profiles



DIFF_KZ_S Vertical Profiles



DIFF_KZ_T Vertical Profiles



Mixing Coefficients Statistics

Active Mixing Regions (coefficient > 1e-6 m²/s)

=====

VISC_AZ: □ EXCELLENT

Median relative error:	0.000000%
Mean relative error:	0.020317%
Max relative error:	210.360572%
95th percentile error:	0.002085%
Median absolute diff:	0.000000e+00 m ² /s
Points compared:	242000

DIFF_KZ_S: □ EXCELLENT

Median relative error:	0.005451%
Mean relative error:	0.301578%
Max relative error:	602.499892%
95th percentile error:	0.184636%
Median absolute diff:	2.380578e-07 m ² /s
Points compared:	19603

DIFF_KZ_T: □ EXCELLENT

Median relative error:	0.005451%
Mean relative error:	0.301578%
Max relative error:	602.499892%
95th percentile error:	0.184636%
Median absolute diff:	2.380578e-07 m ² /s
Points compared:	19603

GHAT: □ EXCELLENT

Median relative error:	0.003249%
Mean relative error:	0.340863%
Max relative error:	416.030265%
Points compared:	13391

Validation Summary & Assessment

OVERALL VALIDATION RESULT:

Status: ☒ NEEDS REVIEW

COMPONENT RESULTS:

HBL (Boundary Layer Depth):	☒ CHECK
Max difference:	20.279218 m
Mean relative error:	0.0604%
VISC_AZ:	☐ EXCELLENT
Median relative error:	0.000000%
DIFF_KZ_S:	☐ EXCELLENT
Median relative error:	0.005451%
DIFF_KZ_T:	☐ EXCELLENT
Median relative error:	0.005451%
GHAT:	☐ EXCELLENT
Median relative error:	0.003249%

PROVENANCE:

UUID Match:	☐ Yes
MITgcm UUID:	9ce2edbc-2bbc-4d49-b1b3-63d37cb0...
Python UUID:	9ce2edbc-2bbc-4d49-b1b3-63d37cb0...

ACCEPTANCE CRITERIA:

☐ EXCELLENT:	HBL < 0.1m, mixing < 0.1% error
☐ GOOD:	HBL < 1.0m, mixing < 1.0% error
☒ CHECK:	Exceeds thresholds - investigate